

Sparsification ladder

Gate placement expands by row; pressure remains an independent intervention.

Step	Ladder ID	One-sided / nonlinearity				Symmetric			Pressure	$R_{\text{model}}^{\text{max}}$ at $T = 2,048$		
		h	m	a	z	q	k	v		17M 14M config	70M	400M 410M config
1	A0	G							none	0.000%	0.000%	0.000%
2	A1-H	R							none	4.278%	12.354%	24.925%
3	A1-H-L1	R							L1N	4.278%	12.354%	24.925%
4	A1-H-OL1	R							OL1	4.278%	12.354%	24.925%
5	A4-Z	+	+	+	+				none	12.833%	37.063%	74.776%
6	A4-Z-OL1	+	+	+	+				OL1	12.833%	37.063%	74.776%
7	A7-Z	+	+	+	+	±	±	±	none	29.952%	49.424%	87.245%
8	A7-Z-OL1	+	+	+	+	±	±	±	OL1	29.952%	49.424%	87.245%

Gate key  GeLU at h  ReLU at h  $G_{+, \kappa}$  $G_{\pm, \kappa}$  identity / no gate

Pressure, when present, is applied to every colored gate site in the row. A7-Z is ladder shorthand for the operational topology A7-Z-POST; q and k are the post-RoPE operands. Pressure-only changes do not alter the topology-conditioned architecture ceiling.

Ceiling assumptions
Full-sequence, uncached Pythia; $d_f = 4d$; dense LM head included. Requested size labels use $(L, d, V) = (6, 128, 50,304)$, $(6, 512, 50,304)$, and $(24, 1,024, 50,304)$, respectively.